Chapter 1: Laws of Motion

Newton's second law states that the net force on an object equals its mass times its acceleration. It explains why heavier objects need more force to speed up.

Chapter 2: Energy

The kinetic energy of a moving object is half its mass multiplied by the square of its speed.

Chapter 3: Photosynthesis

Photosynthesis converts light energy into chemical energy. Chlorophyll absorbs light in the chloroplast.

Chapter 4: Cell Biology

The cell membrane is selectively permeable. Osmosis is the movement of water across a membrane.